Fiscal Year:	FY 2015	**Task Last Updated:**	FY 05/26/2015
PI Name:	Blancaflor, Elison B. Ph.D.		
Project Title:	Utilizing the Advanced Biological Research System (ABRS) on the International Space Station (ISS) to Uncover Microgravity's Impact on Root Development and Cell Wall Architecture		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Bioregenerative Life Support		
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Organization Name:	NASA Kennedy Space Center		
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City:	Kennedy Space Center	**State:**	FL
Zip Code:	32955	**Congressional District:**	8
Comments:	NOTE: Dr. Blancaflor moved to NASA Kennedy Space Center in spring 2021; as a PI, his prior grants were for work at the Noble Foundation in Oklahoma.		
Project Type:	Flight	**Solicitation / Funding Source:**	2011 Space Biology NNH11ZTT002N
Start Date:	07/25/2012	**End Date:**	07/24/2016
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA KSC
Contact Monitor:	Levine, Howard	**Contact Phone:**	321-861-3502
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: Extended to 7/24/2016 per PI and NSSC information (Ed., 5/27/15)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Nakashima, Jin (Noble Foundation) Tang, Yuhong (Noble Foundation)		
Grant/Contract No.:	NNX12AM94G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The overall goal of this proposal is to uncover the underlying molecular mechanisms by which microgravity impacts cell wall architecture in the model plant Arabidopsis thaliana using the Advanced Biological Research System (ABRS) housed on the International Space Station (ISS). This proposal builds on a robust dataset from our rapid turnaround spaceflight experiments, which utilized the Biological Research in Canisters (BRIC16) hardware on the STS-131 mission. Through experiments with the BRIC hardware, we discovered that spaceflight had a profound influence on the expression of genes involved in regulating cell wall architecture. This led us to hypothesize that plants respond to microgravity in large part through the transcriptional reprogramming of genes that control cell wall remodeling. Because of the prominent root developmental phenotypes (i.e., primary root skewing and inhibition of root hair growth) that we observed in our BRIC experiments, it appears that microgravity-dependent changes in cell wall-related transcripts are predominantly manifested in roots. Furthermore, because primary root and root cell wall ultrastructural defects were most pronounced in a mutant to a vegetative actin isoform (act2), we hypothesize that microgravity predominantly affects genes that lie within signaling networks linking the actin cytoskeleton with the plant extracellular matrix. Given our successful experiments with BRIC, we are in an excellent position to effectively utilize the unique capabilities of the ABRS hardware on the ISS to test the aforementioned hypotheses. We will use a new algorithm that we developed from our BRIC16 experiments to more rigorously quantify root growth behavior of wild-type and a higher order vegetative actin mutant (act2 act8 double mutant) (aim 1). This will clarify the extent of actin involvement in specifying root orientation and cell wall remodeling in space. Through the use of monoclonal antibodies and immunocytochemistry, we will also determine how the distribution and abundance of specific cell wall components change in microgravity, and correlate such changes with root developmental responses and modifications in gene expression (aim 2). Finally, we will employ more sensitive deep sequencing technologies (e.g., RNA-Seq) to discover novel microgravity-induced transcriptional changes in plants (aim 3). Knowledge from our spaceflight studies will then be translated into corresponding gene functional studies (e.g., mutant analysis and overexpression), which we expect will lead to the discovery of new players that modulate root development on Earth and in microgravity. The proposed work is therefore relevant to this NASA-NRA because it will address fundamental questions on how gravity modulates the plant's internal cytoarchitecture and the molecular processes that govern the development of its cell walls. This will be significant for NASA's mission as extrapolated in the priorities of the Fundamental Space Biology Plan for 2010-2020 because it will provide us with basic knowledge to guide genetic engineering efforts to generate plants that are better adapted to the weightless environment of spaceflight. This in turn will be beneficial for the design of large plant habitats on the ISS and future space colonies.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research will provide a deeper understanding of how microgravity influences plant development. The focus of this research will be on the development of roots and how they modify the expression of their genes to adapt to the microgravity environment of spaceflight. This research will address one of high priority recommendations of the National Research Council's 2011 decadal survey report, which is to contribute significantly to building a robust spaceflight program to analyze plant physiological responses to stimuli encountered in microgravity. Research areas depicted in this proposal, namely root development and cell wall biology, are significant for NASA's goal of developing large plant habitats on the International Space Station (ISS) and future space colonies. We expect that information gained from basic studies of root growth in microgravity will guide the development of plant varieties better equipped to optimize nutrient and water acquisition, and waste recycling in enclosed regenerative life support systems. Furthermore, because upright plants evolved under the 1-g conditions of Earth in large part through the mechanical strength of its cell walls, this research will provide a deeper mechanistic understanding of the genetic networks that specify how cell walls are constructed in microgravity. In space, such knowledge will enable technologies for designing plants that can sustain optimal growth and stature to overcome the adverse impact that microgravity has on cell wall development. With regard to Earth benefits, understanding of molecular pathways that govern root growth and cell wall development, will benefit agricultural practices and bioenergy research on Earth.
	As noted in our report last year, the original APEX (Advanced Plant EXperiments) proposal submission was designed to use the Advanced Biological Research System (ABRS) on board the ISS. Unfortunately, the ABRS hardware on the ISS experienced technical issues in regard to the control of specific environmental parameters. These issues could not be resolved in time for SpaceX-3 launch, which was the mission we were assigned to. Thus, we were de-manifested from SpaceX-3 and moved to SpaceX-5. Continued technical issues with ABRS also required us to use an alternative hardware called the vegetable production system (VEGGIE) for our experiments. We are pleased to report that SpaceX-5 carrying APEX 03-1 launched on January 10, 2015 after a couple of delays. Experiment activation on-orbit occurred on January 15, 2015. 6 day harvest of seedlings took place on January 21, 2015 and the 11 day harvest was completed on January 26, 2015. The Dragon capsule carrying the Kennedy Space Center (KSC) Fixation Tubes (KFTs) with preserved plant material returned to Earth on February 10, 2015. After de-integration of KFTs at KSC, fixed samples were delivered to the Noble Foundation on February 18, 2015. Inspection and processing of fixed plant material began immediately. For APEX 03-1, we flew ten 9 cm x 9 cm Petri dishes containing 18-24 Arabidopsis seeds. Germination of seeds took place only in orbit. Controlled germination of seeds was accomplished using far-red light seed pretreatment and maintaining plates at 40C in the dark. Experiment activation was triggered on orbit by removing plates from cold stowage and mounting the plates vertically on the VEGGIE unit. Upon exposure to light and 230C temperature on VEGGIE, the seed germinated. Based on the number of plants recovered upon sample return, we found that we had more than 90% germination. No problems with premature germination were reported by the crew. After 6 days and 11 days of experiment activation, seedlings were photographed by ISS commander Butch Wilmore. A corresponding set of ground control seedlings were photographed at the KSC-ISSES facility. For each time point (i.e., 6 days and 11 days), seedlings were carefully transferred to KFTs containing either glutaraldehyde/paraformaldehyde solution or RNALater. The former solution allowed preservation of cell structure for downstream microscopic analysis. The latter solution preserved material for gene expression studies using RNASeq. There were problems encountered in KFT actuation of ground control samples. Most notable were samples from 6 day old ground control seedlings fixed in RNALater. In cases where KFTs failed to actuate, seedlings were not in contact with the RNALater fixative for about 30 minutes. Seedlings in defective KFTs were eventually transferred manually to falcon tubes containing RNALater. However, the 30 minutes that seedlings were exposed to air within the jammed KFTs

Task Progress:	might have triggered a desiccation stress responses. Jammed KFTs also occurred during the 11 day harvest operations. However, because the team was aware of the possibility that some of the newly constructed KFTs might not actuate properly, contingency measures were planned. The advanced planning allowed the team to immediately transfer seedlings from jammed KFTs to falcon tubes containing fixatives that mirrored the time of actuation in space. Therefore, risks associated with jammed KFTs were mitigated. Roots and shoots from RNALater fixed plants (space and corresponding ground controls) were separated manually. RNA from each tissue type was extracted following standard protocols. RNA extraction was completed on March 17 and 18, 2015. We sent RNA to our genomics core facility to determine RNA integrity numbers (RIN). RIN is an indicator of RNA quality and therefore will inform us if plant material obtained from APEX 03-1 will generate reliable data sets. The Bioanalyzer (i.e., the instrument used to determine RIN) algorithm assigns RIN values from a scale of 1 (most degraded) to 10 (least degraded). RIN values from both space samples and the corresponding ground controls ranged from 9.4 to 10. These values indicate that RNA quality from APEX 03-1 material was very high. The high RNA quality now allows us to move forward with library construction for sequencing. This task is currently in progress. Completion of the RNA quality tests is crucial for aim 3 of our proposal. The RNA yield from roots and shoots ranged from 200 ng/microL to 600 ng/microL. The volume of RNA we have on hand is about 40 microL. We therefore have sufficient RNA to complete the proposed RNASeq studies and verification assays by quantitative real-time PCR. The total number of RNA samples we completed for yield and quality analysis is 24 representing 2 time points (6 and 11 days), 2 tissue types (shoots and roots), 2 treatments (space versus ground), and 3 biological replicates. We also completed preliminary processing of seedlings fixed in glutaraldehyde/paraformaldehyde for downstream structural and immunological assays. Individual seedlings are now stored in 15 mL falcon tubes awaiting embedding. For this reporting period we also published a paper on our transcript profiling work on Arabidopsis seedlings grown in the BRIC hardware flown on the second to the last mission of the Space Shuttle Discovery (STS-131) (see Kwon et al., 2015). In this paper we report that genes involved in oxidative stress and cell wall remodeling were differentially regulated in microgravity in both wild-type and actin mutants. Several of the genes associated with cell walls and oxidative stress that were downregulated in microgravity were preferentially expressed in root hairs based on in silico analyses. Reverse genetic screening for mutants to some of the root-hair-expressed genes downregulated in microgravity led to the identification of transfer (T)- DNA insertion lines in two class III peroxidase genes (Valério et al., 2004) with short root hairs due to their tips rupturing. The isolation of new peroxidase mutants that displayed root hair phenotypes coupled with the observation that microgravity grown seedlings had shorter root hairs, supports our spaceflight transcriptomic studies, and provides evidence linking microgravity-induced cell wall and root hair defects with changes in gene expression. These results point to the spacecraft environment as a trigger for oxidative stress responses in plants that can be manifested in part as disrupted cell wall development and defective root hair growth. The PI (Blancaflor) also completed editing a book for Springer title “Plant Gravitropism: Methods and Protocols.” This volume contains a collection of methods for studying gravity responses in a variety of plant organs and cell types. Many of the contributors to this volume are NASA funded PIs including Dr. Jin Nakashima (Co-PI in this proposal). Several chapters in this volume provide overviews on plant space biology.

Bibliography Type:	Description: (Last Updated: 01/13/2026)
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Abstracts for Journals and Proceedings	Sparks JA, Kwon T, Nakashima J, Allen SN, Tang Y, Blancaflor EB. "Transcriptional response of Arabidopsis seedlings to microgravity reveals remodeling of cell wall genes associated with root hair development." Presented at the 30th Annual Meeting of the American Society for Gravitational and Space Research, Pasadena, CA, October 22-26, 2014. 30th Annual Meeting of the American Society for Gravitational and Space Research, Pasadena, CA, October 22-26, 2014. , Oct-2014
Abstracts for Journals and Proceedings	Sparks JA, Kwon T, Nakashima J, Blancaflor EB. "Hypersensitive to Latrunculin B is a Trans-Golgi Network-localized protein involved in cross-talk between actin and endomembranes." Presented at the 30th Annual Meeting of the American Society for Gravitational and Space Research, Pasadena, CA, October 22-26, 2014. Concurrent session 7. Plants-Flight and Ground. 30th Annual Meeting of the American Society for Gravitational and Space Research, Pasadena, CA, October 22-26, 2014. , Oct-2014
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Books/Book Chapters	Blancaflor EB (Ed.) "Plant Gravitropism: Methods and Protocols." Ed. E.B. Blancaflor. New York : Springer, 2015. Methods in Molecular Biology series, vol. 1309. http://dx.doi.org/10.1007/978-1-4939-2697-8 , May-2015